Syllabus covered for Stat-2 course (2023)

- **Statistical Inference**
  - **Testing of Hypothesis**
    - **Introduction**
      - **Definitions**
    - **Tests concerning univariate normal population**
      - **Tests for population mean**
      - **Tests for population variance**
    - **Tests concerning two univariate normal populations**
      - **Tests for differences of population mean**
      - **Tests for ratios of population variance**
    - **Hypothesis testing using R**
    - **Tests concerning bivariate normal distributions**
      - **Comparison of two means**
      - **Test for the correlation coefficient (null case)**
      - **Test for the correlation coefficient (non-null case)**
      - **Tests concerning two variances**
    - **Non-parametric Tests**
      - **Introduction**
      - **Wald-Wolfowitz Run test**
      - **Sign Test**
      - **Wilcoxon Signed Rank test**
      - **Mann-Whitney Test**
    - **Likelihood Ratio Tests**
      - **Introduction**
      - **Properties**
      - **Tests**
      - **Power function**
      - **Ideal test**
    - **More on Hypothesis testing**
      - **Most Powerful Critical Region**
      - **Uniformly Most Powerful Critical Region**
      - **Unbiased Critical Region**
      - **Uniformly Most Powerful Unbiased Critical Region**
      - **Neyman Pearson Lemma**
      - **Power function and its use**
    - **ANOVA**
- **Measurement Scales**
- **Association of attributes**
  - **Introduction**
  - **Independence**
  - **Association**

- **Yule's first measure of association**
- **Yule's second measure of association**
- **Odds ratio**
- **Chi-squared Tests**
    - **Test for goodness of fit**
    - **Test for independence of two attributes**